

The due date for submitting this assignment has passed.

Due on 2026-02-25, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-02-25, 08:53 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

### Multiple Select Questions (MSQ)

1) In a particular year, the profit (in lakhs of ₹) of Star Fish company is given by the polynomial **1 point**  
 $P(x) = ax^2 + bx + c$  where  $x$  denotes the number of months since the beginning of the year (i.e.,  $x = 1$  denotes January,  $x = 2$  denotes February, and so on). In January and February the company made a loss of ₹45(in lakhs), and ₹19(in lakhs) respectively, and in March the company made a profit of ₹3(in lakhs). Let the loss be represented by negative of profit. Choose the correct set of options based on the given information.

- ☐ The maximum profit will be in the month of May.
- ☒ The maximum profit will be in the month of August.
- ☒ The maximum monthly profit amount is ₹53 lakh.
- ☐ The maximum monthly profit amount is ₹35 lakh.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The maximum profit will be in the month of August.

The maximum monthly profit amount is ₹53 lakh.

2) If  $A$  be a  $3 \times 4$  matrix and  $b$  be a  $3 \times 1$  matrix, then choose the set of correct options.

**1 point**

- If  $(A|b)$  be the augmented matrix and  $(A'|b')$  be the matrix obtained from  $(A|b)$  after a finite number of elementary row operations then the system  $Ax = b$  and the system  $A'x = b'$  have the same set of solutions.
- ☒ a finite number of elementary row operations then the system  $Ax = b$  and the system  $A'x = b'$  have the same set of solutions.

- ☐ If  $(A' | b')$  is the reduced row echelon form of  $(A | b)$  then the system  $A'x = b'$  has at least one solution.
- ☒ If  $(A' | b')$  is the reduced row echelon form of  $(A | b)$ , then  $A'$  is also in reduced row echelon form.
- ☒ If  $(A' | b')$  is the reduced row echelon form of  $(A | b)$  and there is no row such that the only non zero entry lies in the last column of  $(A' | b')$  then the system  $Ax = b$  has at least one solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If  $(A | b)$  be the augmented matrix and  $(A' | b')$  be the matrix obtained from  $(A | b)$  after a finite number of elementary row operations then the system  $Ax = b$  and the system  $A'x = b'$  have the same set of solutions.

If  $(A' | b')$  is the reduced row echelon form of  $(A | b)$ , then  $A'$  is also in reduced row echelon form.

If  $(A' | b')$  is the reduced row echelon form of  $(A | b)$  and there is no row such that the only non zero entry lies in the last column of  $(A' | b')$  then the system  $Ax = b$  has at least one solution.

3) Choose the set of correct options

**1 point**

- ☒ If the sum of all the elements of each row of a matrix  $A$  is 0, then  $A$  is not invertible.
- ☒ If  $E$  is a matrix of order  $3 \times 3$  obtained from the identity matrix by a finite number of elementary row operations then  $E$  is invertible.
- ☐ Any system of linear equations has at least one solution.
- ☐ If  $A$  is a matrix of order  $3 \times 3$  and  $\det(A) = 3$  then  $\det(\text{Adj}(A)) = 3$ .
- ☒ If  $A$  is a matrix of order  $3 \times 3$  and  $\det(A) = 3$  then  $\det(\text{Adj}(A)) = 9$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

If the sum of all the elements of each row of a matrix  $A$  is 0, then  $A$  is not invertible.

If  $E$  is a matrix of order  $3 \times 3$  obtained from the identity matrix by a finite number of elementary row operations then  $E$  is invertible.

If  $A$  is a matrix of order  $3 \times 3$  and  $\det(A) = 3$  then  $\det(\text{Adj}(A)) = 9$ .

4) Ramya bought 1 comic book, 2 horror books, and 1 novel from a bookshop which cost her ₹1000. Romy bought 2 comic books, 5 horror books, and 1 novel which cost him ₹2000. Farjana bought 4 comic books, 5 horror books, and  $c$  novels from a shop which cost her ₹ $d$ . If  $x_1, x_2$ , and  $x_3$  represent the price of each comic book, horror book, and novel, respectively, then choose the set of correct options. **1 point**

☒ The matrix representation to find  $x_1, x_2$  and  $x_3$  is  $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 4 & 5 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1000 \\ 2000 \\ d \end{bmatrix}$

☒ The matrix representation to find  $x_1, x_2$  and  $x_3$  is  $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$

☐ The matrix representation to find  $x_1, x_2$  and  $x_3$  is  $\begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$

☐ If Farjana tries to find  $x_1, x_2$ , and  $x_3$  using appropriate matrix representation by taking  $c = 2$  and  $d = 4000$ , then the price of each comic book that she thus arrives at, will not be unique.

☐ If  $c = 7$  and  $d = 4000$ , then the price of each comic book cannot be determined from this data.

☒ If  $c = 7$  and  $d = 3000$ , then the shopkeeper has made a mistake.

☒ If  $c = 2$  and  $d = 3000$ , then the price of each comic book can be determined from the data.

Partially Correct.

Score: 0.8

Accepted Answers:

The matrix representation to find  $x_1, x_2$  and  $x_3$  is  $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 4 & 5 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1000 \\ 2000 \\ d \end{bmatrix}$

The matrix representation to find  $x_1, x_2$  and  $x_3$  is  $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$

If  $c = 7$  and  $d = 4000$ , then the price of each comic book cannot be determined from this data.

If  $c = 7$  and  $d = 3000$ , then the shopkeeper has made a mistake.

If  $c = 2$  and  $d = 3000$ , then the price of each comic book can be determined from the data.

5) Let  $A$  be an  $m \times n$  matrix such that  $m < n$ . How many solutions does  $Ax = 0$  have?

**1 point**

- ☐ Exactly one solution.
- ☒ Infinitely many solutions.
- ☐ No solution.
- ☐ Insufficient data. This depends on the entries of  $A$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

Infinitely many solutions.

6) Let  $A$  be an  $n \times n$  matrix such that  $\sum_{j=1}^n a_{ij} = 0$  for all  $i$ . How many solutions does  $A\mathbf{x} = \mathbf{0}$  have?

**1 point**

- ☐ Exactly one solution.
- ☒ Infinitely many solutions.
- ☐ No solution.
- ☐ Insufficient data. This depends on the entries of  $A$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

Infinitely many solutions.

**Numerical Answer Type (NAT):**

7) How many solutions does the following system have? Enter 1 if there is only one solution, 0 if there are no solutions and -1 if there are infinitely many solutions. The coefficient matrix  $A$  and the vector  $b$  are given below:

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

-1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -1

**1 point**

8) Let the reduced row echelon form of a matrix  $A$  be  $\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{6} \\ 0 & 0 & 1 & \frac{1}{6} \end{bmatrix}$ . The first, second, third and fourth columns of  $A$  are  $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ ,  $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ ,  $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$  and  $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ , respectively. The value of  $a + b + c$  is

-12

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

**1 point**

9)  $A$  is the reduced row echelon form of the matrix  $\begin{bmatrix} 1 & 3 & 0 & 0 \\ 4 & 1 & 5 & 5 \\ 2 & 2 & 7 & 91 \\ 3 & 9 & 0 & 0 \end{bmatrix}$ . Then determinant of  $A$  is

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

**1 point**

10) If  $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$  is a solution of the system of equations

$$7x + 2y + z = 8$$

$$3y - z = 2, \text{ then the value of } x + y + z \text{ is}$$

$$-3x + 4y - 2z = 5$$

121

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 121

**1 point**

11) Let  $A = [1 \ 7 \ 2 \ 9]$  and  $M$  denote the reduced row echelon form of  $A^T A$ . The number of non-zero rows of  $M$  is

1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1

**1 point**

12) Consider the curve corresponding to the function  $f(x) = ax^3 + bx^2 + cx + d$ . The following points lie on the curve:  $(1, 1)$ ,  $(2, 7)$ ,  $(-1, -5)$ ,  $(3, 23)$ . Find the value of  $a - b + c - d$ .

5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 5

**1 point**

### Comprehension Type Question:

The network in Figure: M2W2GA1 shows a proposed plan for flow of traffic around a park. All the streets are assumed to be one-way and the arrows denote the direction of flow of traffic. The plan calls for a computerized traffic light at the South Street. Let  $2x_1$ ,  $3x_2$ ,  $2x_3$ , and  $x_4$  denote the average number (per hour) of vehicles expected to pass through the connecting streets (e.g.,  $2x_1$  denote the average number (per hour) of vehicles expected to pass through the street connecting the North Street and West Street as shown in Figure: M2W2GA1). 400, 1000, 900, and  $c$  denote the average number (per hour) of vehicles expected to pass through West, North, East, and South Streets respectively.

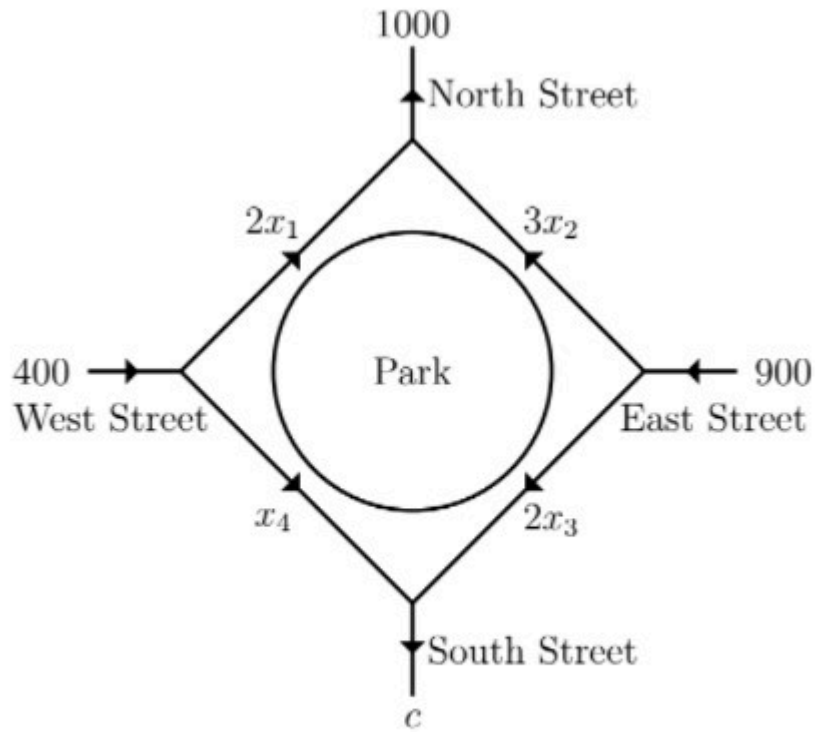


Figure: M2W2GA1

13) Which of the following options are correct?

**1 point**

The system of equations corresponding to the flow of expected traffic according to the given data above, will be



$$2x_1 + 3x_2 = 1000$$

$$3x_2 + 2x_3 = 900$$

$$2x_3 + x_4 = c$$

$$2x_1 + x_4 = 400$$

The system of equations corresponding to the flow of expected traffic according to the given data above, will be



$$2x_1 + 3x_2 = 900$$

$$3x_2 + 2x_3 = 1000$$

$$2x_3 + x_4 = 400$$

$$2x_1 + x_4 = c$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is





$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 900 \\ 1000 \\ 400 \\ c \end{bmatrix}$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is



$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

Partially Correct.

Score: 0.5

Accepted Answers:

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

$$2x_1 + 3x_2 = 1000$$

$$3x_2 + 2x_3 = 900$$

$$2x_3 + x_4 = c$$

$$2x_1 + x_4 = 400$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

14) How many vehicles are expected to pass through the South Street per hour on an average?

300

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 300

**1 point**

15) Match the names of the street in Column A with the maximum and minimum number of vehicles expected to pass through the street on an average (per hour) in Column B and Column C, respectively; in Table M2W2GA1. **1 point**

|    | Name of the connecting street    |      | The maximum number of vehicles expected to pass through the street (per hour) |    | The minimum number of vehicles expected to pass through the street (per hour) |
|----|----------------------------------|------|-------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------|
|    | Column A                         |      | Column B                                                                      |    | Column C                                                                      |
| a) | Connecting West and North street | i)   | 300                                                                           | 1) | 0                                                                             |
| b) | Connecting East and North street | ii)  | 300                                                                           | 2) | 100                                                                           |
| c) | Connecting East and South street | iii) | 400                                                                           | 3) | 600                                                                           |
| d) | Connecting West and South street | iv)  | 900                                                                           | 4) | 0                                                                             |

Table : M2W2GA1

- ☐ d → i → 2
- ☐ b → iii → 3.
- ☒ b → iv → 3.
- ☒ d → ii → 1.
- ☒ a → iii → 2.
- ☐ a → iii → 4.
- ☒ c → i → 4.

Yes, the answer is correct.

Score: 1

Accepted Answers:

- b → iv → 3.
- d → ii → 1.
- a → iii → 2.
- c → i → 4.

**NOTE FOR OPTION-2 IN QUESTION-3: [For elementary row operation of type-2 (scaling rows by a constant), the constant has to be non-zero. Scaling a row by zero is not permitted] Please answer this question keeping this in mind.**